

■ HarvestIQ Ghana

A Smarter Way to Farm

How Technology Can Help Ghana's Farmers Grow More, Plan Better, and Know When Their Harvest Is Coming

Research Team

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A Story Every Farmer Knows

Imagine it is March. The rains are coming. You have one hectare of land and enough money to buy seed for one crop. You need to decide: maize or soybean? Your neighbour planted maize last year and had a good harvest. But your soil is different from your neighbour's. Your field is in a different part of the region. And last year's rains are not the same as this year's.

So you plant what your father planted. And you wait. Three months later, you are not sure when the harvest will come. A buyer calls and asks how much produce you will have ready and when. You give your best guess. But it is still just a guess.

This is the reality for millions of smallholder farmers across Ghana. They are making big decisions — decisions that affect their families, their income, and their future — without the information they need to make those decisions well.

HarvestIQ Ghana was built to change that.

Using data from farms across 10 Ghanaian regions — including information about soil quality, rainfall, crop type, and planting dates — HarvestIQ Ghana uses a computer programme to answer two of the most important questions a farmer can ask:

How much crop will I harvest from my farm? When will my harvest be ready?

What HarvestIQ Ghana Does

HarvestIQ Ghana looks at information that already exists about your farm — your soil nutrients, how much rain your area receives, what crop you are growing, and when you planted it — and uses that information to make predictions.

Think of it like a very experienced agriculture officer who has studied thousands of farms across Ghana and can look at your farm's details and tell you what to expect — except it is a computer programme that gives you this answer in seconds.

Your Question	What HarvestIQ Ghana Tells You
How much will I harvest?	An estimate of your yield in kilograms per hectare, based on your soil nutrients and the rainfall your area receives.
When will my harvest be ready?	An estimate of how many days from your planting date to harvest, so you can plan your season and give your crops the best care.

Who Benefits

Who	How They Benefit
Farmers	Know what to expect from their harvest before it happens. Plan the season with more confidence. Give buyers accurate
Buyers / Traders	Know how much produce will be available from different regions and when, so they can plan purchases and logistics
Extension Officers & NGOs	Understand which soil inputs — nitrogen, phosphorus, potassium — matter most for each crop in each region. Use o

What Information Does It Use?

HarvestIQ Ghana was trained on records from 2,060 farms spread across 10 regions of Ghana. Here is the information it uses for each farm, and why each piece matters:

Information	What It Means	Why It Matters
Nitrogen (%)	Soil nitrogen content	Feeds plant growth and leaf development. Too little means stunted plants and lower yield.
Phosphorus (ppm)	Phosphorus level in soil	Helps roots grow strong and supports flowering and fruiting.
Potassium (ppm)	Potassium level in soil	Builds disease resistance and improves harvest quality and weight.
Soil pH	Acidity or alkalinity	Different crops thrive at different pH levels; wrong pH limits nutrient absorption.
Rainfall (mm)	Seasonal rainfall amount	Too little causes drought stress; too much can cause waterlogging and disease.
Crop type	What crop is being grown	Different crops have completely different yield sizes and growth timelines.
Planting date	When the crop went in	Planting in the right season (Major or Minor) affects maturity time.
Region	Which part of Ghana	Rainfall patterns, temperatures, and soil types vary significantly by region.

Crops included: Cassava · Yam · Maize · Rice · Soybean · Groundnut · Cocoa · Plantain

What We Found

After studying thousands of farm records, some clear and consistent patterns emerged:

- **Soil nutrients matter — a lot.** The three most important factors for yield are nitrogen, phosphorus, and potassium. Farms with healthier nutrient levels consistently produced higher yields across all eight crop types.
- **The crop you choose determines your harvest timeline.** Maize takes roughly 90–120 days. Cassava can take 8–12 months. Soybean typically matures in about 100 days. The model learns these patterns.
- **When you plant makes a real difference.** Ghana's Major Season (March–August) and Minor Season (September–February) both affect how long a crop takes to mature and how much it produces.
- **Rainfall supports yield — but there is a limit.** Extremely high rainfall does not always produce the highest yields. Waterlogging and nutrient washout can reduce yield even in high-rainfall seasons.

What This Tool Cannot Do (Yet)

HarvestIQ Ghana is a first step, not a finished product. Being honest about its limitations is important. Here is what the tool does not yet know:

- Whether you applied fertiliser, how much, and when — this has one of the biggest effects on yield.
- Whether your farm is irrigated or depends only on rainfall.
- The specific variety of seed you planted — improved varieties can yield significantly more than local varieties.
- Whether your crop was affected by pests or disease during the growing season.

Because this information is missing from the data, the model's predictions are estimates — useful guides, not guarantees. The goal for the next version of HarvestIQ Ghana is to include this missing information so that predictions become more accurate and more personalised to each farmer's real situation on the ground.

What Comes Next

- **Collect real farm data** — Work with MoFA Ghana and local agricultural networks to gather verified, district-level farm records.
- **Build a simple mobile app** — Enter soil readings, crop, and planting date in the field and get a prediction instantly. No laptop required.
- **Add fertiliser, irrigation, and seed variety** — Include the inputs that currently limit the model's accuracy.
- **Expand to crop quality prediction** — After harvest, help farmers and buyers know whether produce meets market standards before it leaves the farm.

"The best time to give a farmer data was before planting. The second best time is now."

About the Team

HarvestIQ Ghana was developed collaboratively by a team of six researchers and ML practitioners committed to building data-driven tools for African farmers.

Name	Role / Background
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This project is part of a broader portfolio of Ghana-focused ML work including DumsorData (power outage prediction), HarvestIQ Post-Harvest (crop quality grading), and FoodQwik (AI-powered food ordering).

